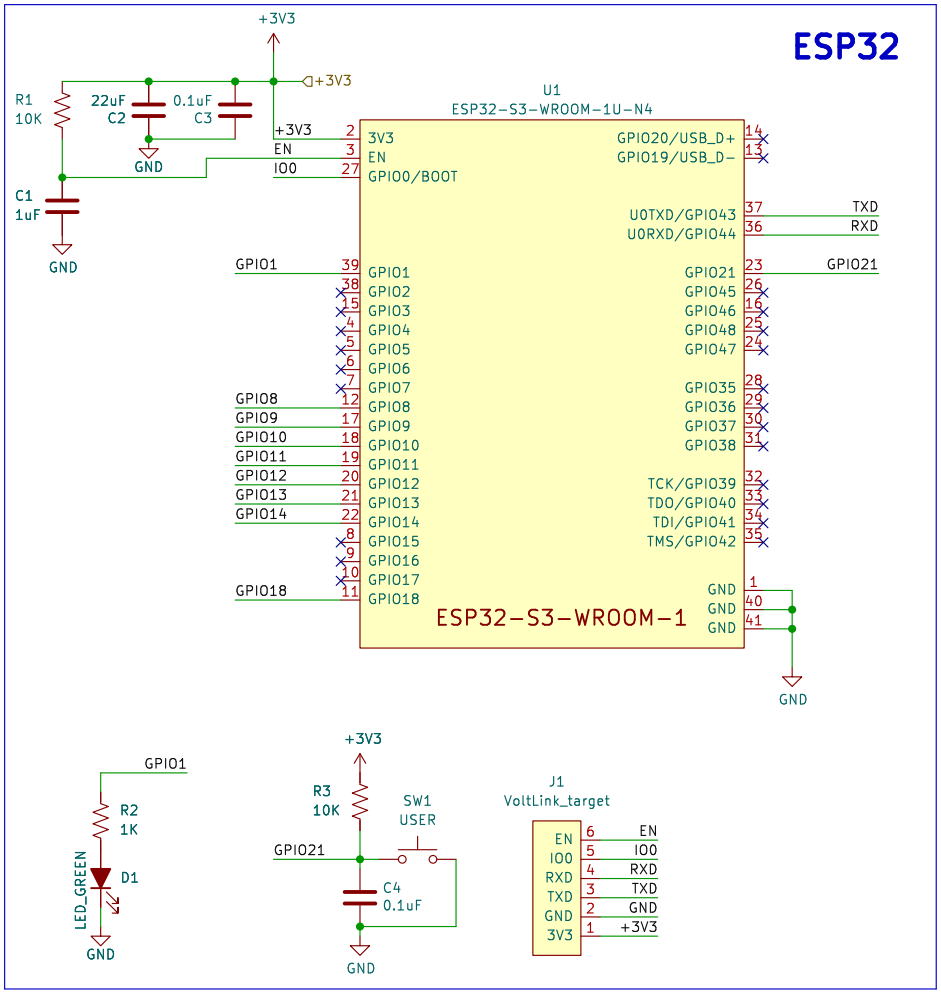
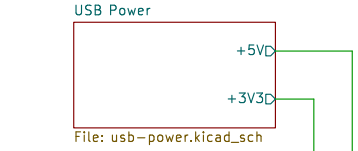
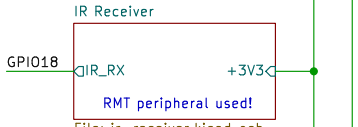
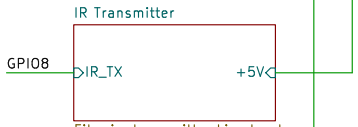
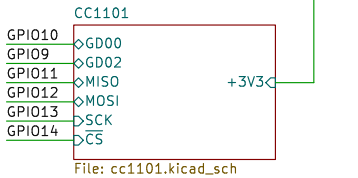
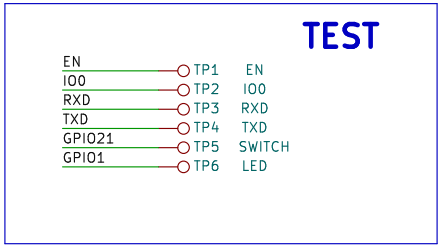




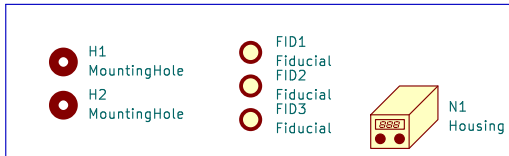








CHANGELOG
RevA: Initial release, based on IRBlaster revB
moved LEDs to center of PCBs, added test points, removed USB data connection, optimized DFM, added RF module.



Strapping pins
- BOOT = IO0
- JTAG = IO3
- VSPI = IO45
- LOG = IO46

SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses. When using Octal Flash or Octal PSRAM or both, GPIO33~37 are connected to SPIIO4 ~ SPIIO7 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO33~37 are also not recommended for other uses.

For 2MB PSRAM version (embedded chip ESP32-S3R2 quad SPI) pins IO35, IO36, IO37 should be free to use. For 8MB PSRAM version (embedded chip ESP32-S3R8 octal SPI) pins IO35, IO36, IO37 connect to PSRAM and are not available for other uses.

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs, USB-JTAG will be disabled by the drivers.

All resistors are 1% unless otherwise mentioned.

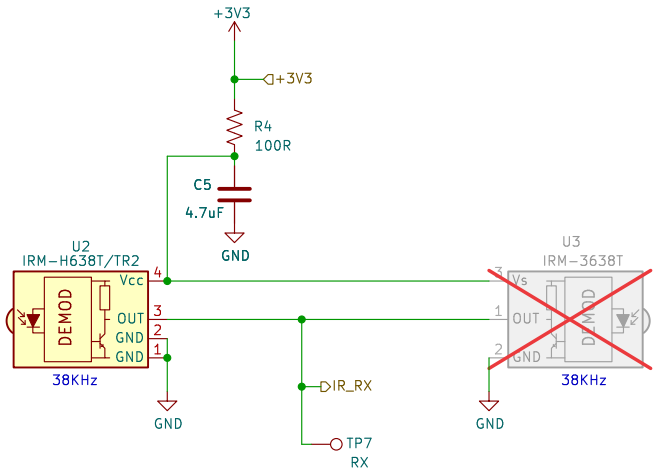
Ezlo

Sheet: /
File: irblaster.kicad_sch

Title: IR+RF Blaster

Size: A4	Date: 2023-03-21	**Rev: A**
KiCad E.D.A. kicad (7.0.0)		Id: 1/5

IR Receiver



Everlight Elec IRM-H638T/TR2
NEC YES
RC5_Philips YES
RC6_Philips YES
RCA_Thomson YES
Toshiba YES
SHarp YES
Sony 12Bit YES
Sony 15Bit NO
Sony 20Bit NO
Matushita YES
Mitsubishi NO
Zenith YES
JVC NO
Continous Code NO
High Data Rate Code NO
*RC6 is only compatible if the data low time is 25ms or more.

Populate U3 OR U4 depending on availability!

All resistors are 1% unless otherwise mentioned.

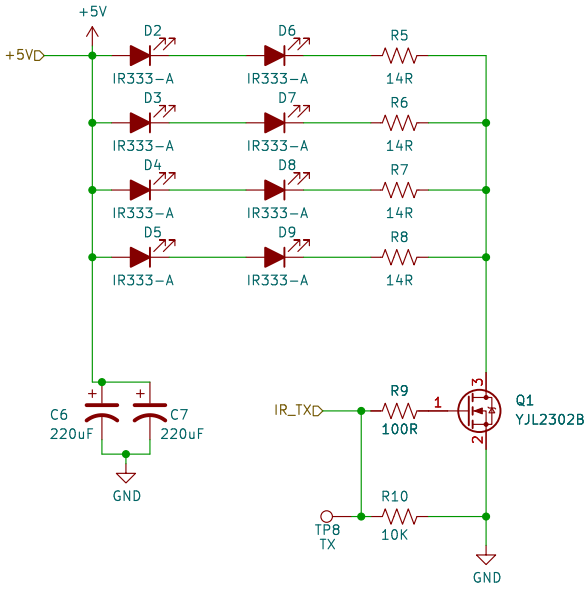
Ezlo

Sheet: /IR Receiver/
File: ir-receiver.kicad_sch

Title: IR+RF Blaster

Size: A4	Date: 2023-03-21	**Rev: A**
KiCad E.D.A. kicad (7.0.0)		Id: 2/5

IR Transmit



940nm If=100mA Vf=1.4V
IR333-A has an angle of 20deg
IR333-C has an angle of 30deg

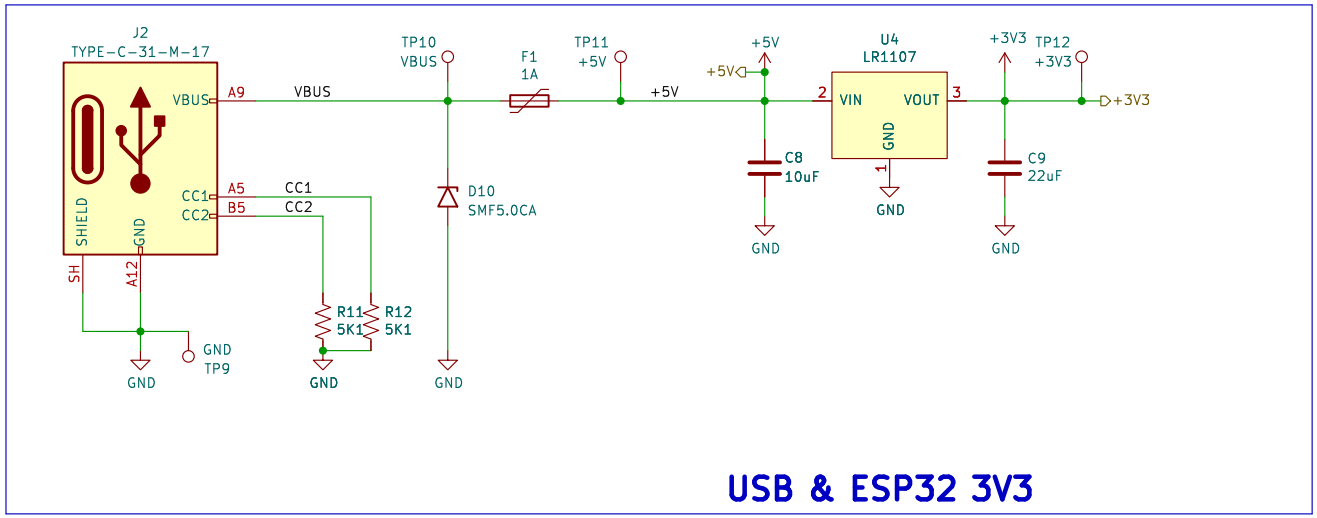
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /IR Transmitter/
File: ir-transmitter.kicad_sch

Title: IR+RF Blaster

Size: A4	Date: 2023-03-21	**Rev: A**
KiCad E.D.A. kicad (7.0.0)		Id: 3/5



All resistors are 1% unless otherwise mentioned.

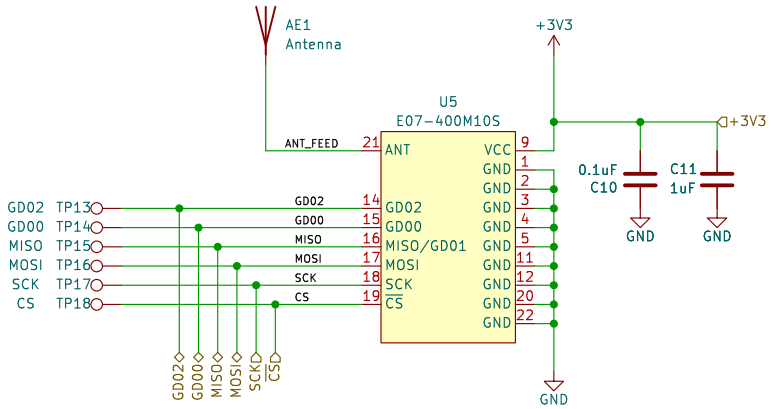
Ezlo

Sheet: /USB Power/
File: usb-power.kicad_sch

Title: IR+RF Blaster

Size: A4	Date: 2023-03-21	**Rev: A**
KiCad E.D.A. kicad (7.0.0)		Id: 4/5

CC1101 RF Blaster



EBYTE module are available in two variants 400MHz and 900MHz
E07-900M10S(has CE and FCC) E07-400M10S(just CE)
For future revisions we should consider implementing our own RF section based on CC1101 or the newer CC110L

All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /CC1101/
File: cc1101.kicad_sch

Title: IR+RF Blaster

Size: A4	Date: 2023-03-21	**Rev: A**
KiCad E.D.A. kicad (7.0.0)		Id: 5/5